Md Hasibul Amin

Software Engineer – Distributed Systems, Cloud Infrastructure & AI/ML

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Software Engineer with 10 years of experience building large-scale distributed systems and cloud infrastructure on Azure and AWS. Experienced in software design and architecture, Kafka/CDC, multi-tenant APIs, CQRS, Kubernetes, CI/CD, and production reliability. Previously led a 12-engineer cross-functional team building systems serving 400K+ daily requests and 5K+ concurrent users, with additional AI/ML research experience in NLP, continual learning, model reliability, and medical imaging.

Professional Experience

Software Engineer July 2023 - Present

Crane Worldwide Logistics LLC *Houston, TX – Global 3PL logistics provider*

- Designed and built from scratch a **resilient Kafka-based billing platform** automating invoice validation, document sequencing/assembly, downstream delivery, and notifications, with idempotent processing, retry policies, and **DLQ-backed** recovery; **99%** of requests complete within **90 seconds**.
- Built **Kafka CDC pipelines** that synchronized tens of millions of production records and propagate ongoing changes within a **30-second latency SLA**.
- Designed **CQRS-based** PostgreSQL read models to offload read-heavy queries from Oracle, improving query response time by **15–18%** and reducing primary-database load.
- Serve as production **DRI** across mission-critical platforms, **triaging** and **debugging** service issues, **leading incident** response and **RCA**.
- Lead **design**, **architecture**, and code reviews with cross-team stakeholders for C#/ASP.NET and F# services and APIs across **AWS** and **Azure**.
- Designed **secure multi-tenant** transportation and warehouse APIs with configuration-driven behavior, **database-per-client isolation**, and least-privilege **RBAC**.
- Own **CI/CD** production releases across **AWS and Azure**, including **Terragrunt** and **Bicep** infrastructure-as-code changes.
- Integrated AI coding assistants – Claude Code and GitHub Copilot with **MCP workflows** across requirements analysis, coding, automated testing, log-driven debugging, and code review.

Lead Software Engineer Sep 2019 - Jul 2021

Truck Lagbe Ltd *Dhaka, Bangladesh – Digital freight marketplace for shippers and truck drivers*

- **Led and mentored** a 12-engineer cross-functional team through code reviews and design guidance, while owning backend and database architecture for a marketplace handling **400K+** daily requests and **5K+** concurrent users.
- Built a **multilingual geo-search engine** mapping informal landmarks to formal addresses using a **trie, inverted index, grid-based spatial hashing,** and f**uzzy matching on a custom Bangla dataset**; achieved 95–97% accuracy and approximately $35K in annual savings.
- Designed **high-availability database architecture** using **CQRS**, replication, **age-based trip partitioning**, caching, and recovery workflows to reduce primary-database load.
- **Architected the core backend and reusable Node.js framework** standardizing APIs, security, logging, and service development across teams.
- Built an **adaptive pricing engine** for fixed trip quotes using distance, day/time, demand, weather, vehicle type, and historical trip data; **A/B-style rollout testing** increased successful bookings by **20–23%**.
- Migrated production workloads to AWS/Kubernetes, improving API response time by **20%** and reducing deployments from 2–3 hours to 20–30 minutes.
- Partnered with Product and third-party payment providers on requirements and architecture, delivering integrations in **Java Spring**.
- Served as Acting CTO for two months, setting engineering priorities and leading a major cross-platform release.

Associate Senior Software Engineer Feb 2017 - Aug 2019

CSL Software Resources Ltd *Dhaka, Bangladesh*

- Built a reusable **C++ access-control library** for C#/ASP.NET applications, reducing development and maintenance effort by **10%.**
- Developed ERP and POS platforms used by **400+** companies and **250+** retail outlets.
- Optimized APIs and SQL workloads, improving dashboard response time by **35%**.

Software Engineer Jan 2015 - Jan 2016

GOBD (Startup) *Dhaka, Bangladesh*

- Built **route-optimization logic** combining live tracking, traffic-signal, mobile-app, and third-party location data, weighted by source priority to select alternate routes.

Research Experience

Independent AI/ML Research Collaborator Late 2024 – Present

Conducted alongside full-time engineering role, with multi-institution co-authors

- Develop and evaluate **continual-learning** methods in Python/PyTorch to reduce **catastrophic forgetting**, with a focus on medical imaging.
- Identified a readout-dependent failure mode that reverses the measured benefit of knowledge distillation in controlled multi-seed studies.
- **Co-authored published research in breast-cancer segmentation** and medical LLM fine-tuning, including evaluation on 614 mammograms and 47K+ QA pairs.
- Built a BERT-based NLP pipeline for COVID-19 vaccine conspiracy and misinformation detection, incorporating Google Perspective API signals for content-quality analysis.

Graduate Research Assistant Sep 2021 – May 2023

University of Houston – Imaging Innovations Laboratory *Houston, TX — During M.S. Computer Science*

- Built an automated **GPU-accelerated CT acquisition and reconstruction pipeline** using **Python** with 360° rotation and precise positional control, reducing end-to-end processing time by **35–40%**.

Selected Publications

- **Under Review**: *Readout-Induced Reversal of Component Attribution: A Controlled Analysis of Replay-Based Continual Learning*
- **IEEE CICN 2026**: *Boundary-Aware Deep Learning Framework for Breast Cancer Image Segmentation: Enhancing Tumor Delineation through BIOU-Driven Evaluation*[IEEE Xplore]
- **IEEE CICN 2026**: *Evaluating Parameter-Efficient Fine-Tuning of Open-Weight LLMs for Medical Question Answering* [IEEE Xplore]
- **Preprints & Artifacts**: *Detecting Conspiracy Theory Against COVID-19 Vaccines [arXiv]* | *Failure Modes of Always-On Inter-Cluster Repulsion in Replay-Based Continual Learning* [arXiv]

Key Skills

Languages: Python, C#, F#, C++, Java, JavaScript, SQL

Backend & Distributed Systems: Kafka, CDC, Microservices, CQRS, REST APIs, Event-Driven Architecture, Workflow Orchestration, Large-Scale Infrastructure, Software Design & Architecture, Multi-Tenant Systems, High Availability, Resiliency, Idempotency, Retry/DLQ, System Design, Object-Oriented Design

Cloud & Platform: AWS, Azure, Kubernetes, Docker, Terraform/Terragrunt, Bicep, Lambda, Azure Functions, CI/CD, Git

Data & Observability: PostgreSQL, Oracle, SQL Server, NoSQL, MongoDB, Redis, OpenTelemetry, Distributed Tracing, Structured Logging, Incident Response, Root Cause Analysis (RCA)

Frameworks: ASP.NET Core/MVC, Node.js, Java Spring

AI/ML & Search: PyTorch, TensorFlow, Continual Learning, NLP, BERT, Information Retrieval, Trie, Inverted Index, Spatial Hashing, Fuzzy Matching, LLMs, RAG, Model Evaluation, A/B Testing, OpenCV, GitHub Copilot, Claude Code, MCP

Education

- **Master of Science in Computer Science** – University of Houston, 2023 | Machine Learning, Deep Learning, NLP, Cloud Computing, Digital Image Processing
- **Bachelor of Science in Software Engineering** – American International University-Bangladesh (AIUB), 2015 | Algorithms, Data Structures, OOP, C++, C#, Java
- **Certifications:** Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | Architecting on AWS